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Detection of Red Palm Weevil Infestation in Coconut Trees Using AI-Based Acoustic Signal Analysis

Team IoTrix

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July 3, 2026

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1 Problem Description and Validation

Problem

Red palm weevil infestation is one of the most critical challenges in coconut cultivation due to its hidden and highly destructive nature. The pest attacks the internal tissues of the coconut palm, particularly the apical bud (the “heart” of the tree), which is essential for growth and survival. Because the infestation develops inside the trunk, early-stage damage is extremely difficult to detect using visual inspection. Although monitoring devices and detection methods are currently used in some farms, these systems often identify the infestation only after significant internal damage has already occurred. By the time clear external symptoms appear—such as audible larval feeding sounds or the oozing of fermented brown fluid—the tree is usually beyond recovery. This delay in detection results in irreversible damage, leading to the loss of coconut palms that require several years of growth and substantial financial investment before reaching the fruit-bearing stage. Therefore, there is a clear need to improve early detection methods to identify infestations at a much earlier stage.



(a) A Red Palm Weevil Larva



(b) An infected tree trunk

Problem Validation

To validate this problem, several field visits were conducted at active coconut farms, where farmers consistently reported red palm weevil (RPW) damage as a major cause of tree loss. These observations were further supported by agricultural experts and direct communication with the Coconut Research Institute of Sri Lanka, confirming that RPW infestation remains one of the country’s top threats to coconut cultivation.

While existing acoustic detection devices provide some level of monitoring, the users themselves highlighted significant operational difficulties during our field interviews. Currently, farmers must rely on listening to RPW sounds through earphones—a process heavily disrupted by outdoor ambient noise, making accurate detection highly challenging. Furthermore, farmers expressed that continuously wearing earphones while working in the field is incredibly inconvenient.



(a) Existing device for RPW detection



(b) Examining the existing device



Figure 3: One of the meetings with Coconut Estate Owners - Mr. Nilantha Jayasuriya.

2 Proposed Solution

To overcome these barriers, providing a real-time visual output through an integrated display would be highly beneficial. Transitioning from purely audio-based monitoring to a clear display directly addresses user feedback, eliminating the strain of noise interference and the discomfort of constant earphone usage, thereby drastically improving the feasibility of early, effective infestation detection.

We propose an AI-integrated, human-centric, low-cost device for the early detection of red palm weevil infestations in coconut trees. The system employs a sensitive probe inserted into the trunk to capture subtle internal acoustic signals produced by larval activity, enabling detection at an early

stage that is otherwise imperceptible to the human ear. These signals are filtered to eliminate environmental noise and analyzed using AI/ML algorithms to generate a clear and interpretable probability score indicating the likelihood of infestation. By transforming complex acoustic data into simple, actionable insights, the device supports informed human decision-making while reducing reliance on risky manual inspections. Its cost-effective design ensures accessibility for smallholder farmers in rural communities, improves crop protection and productivity, enhances worker safety, and combines machine intelligence with human judgment.

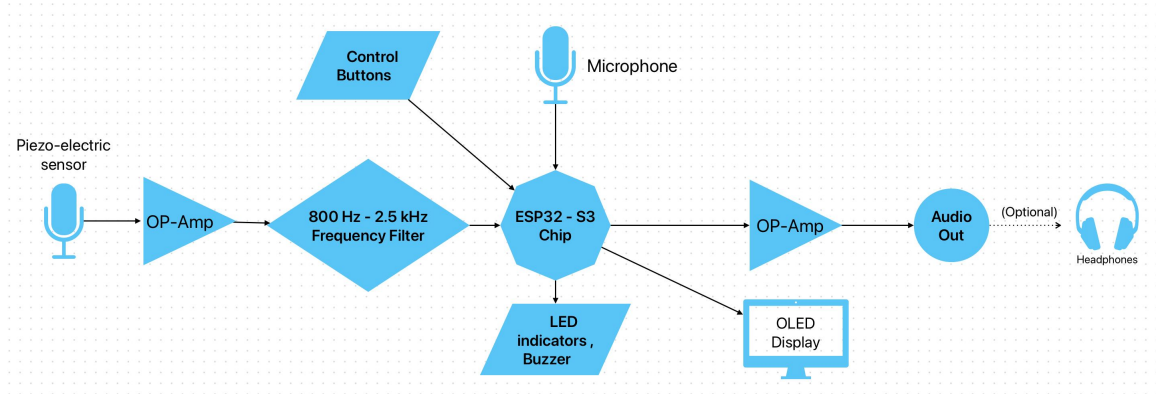


Figure 4: Solution Architecture

3 Product Design and Architecture

3.1 Printed Circuit Board

The custom printed circuit board is designed as a low-power, edge-computing node centered around the ESP32-WROOM-32E microcontroller.

- Processing Core

The ESP32 handles both the digital signal processing of incoming audio and the execution of the onboard machine learning model, taking advantage of its efficient architecture and native wireless capabilities.

- Analog Front-End

The board features a dedicated high-sensitivity acoustic monitor circuit. It uses a 3.5mm audio jack to interface with the external acoustic probe, feeding the raw signal through an operational amplifier stage for signal conditioning and noise reduction before it reaches the microcontroller's ADC.

- Power and Interfacing

Power regulation is managed by an AMS1117-3.3 LDO, safely stepping down a 6V external supply to the required 3.3V logic level. A CP2102 USB-to-UART bridge is integrated for seamless programming, serial debugging, and data extraction.

- User I/O

The board includes standard tactile buttons for system resets and triggering the analysis sequence, alongside dedicated red and green LED indicators for immediate visual feedback on the device's status.

3.2 Enclosure

The physical enclosure is engineered to prioritize field usability, ensuring the internal electronics are protected while giving the operator intuitive control over the acoustic analysis process.

- Operator Control Panel

The faceplate features a clean, functional layout with a dedicated "Analyze" push-button to manually trigger the inference cycle and a standard ON/OFF slide switch for power management.

- Auditory Monitoring

To allow farmers or technicians to manually verify the acoustic signals, the design includes a headphone jack and a rotary dial for adjusting the sensor's listening sensitivity or volume.

- Visual Feedback

A designated rectangular cutout allows for the integration of a digital display (such as an OLED or LCD screen), which can provide real-time system metrics, battery life, and the final diagnostic output directly on the device.

3.3 TinyML Model

The software intelligence relies on an ultra-lightweight, edge-optimized quantized (int8) neural network trained to classify the specific acoustic signatures of RPW larvae.

- Digital Signal Processing (DSP)

Raw audio is pre-processed using a Mel-filterbank energy feature extraction pipeline. It utilizes a frame length of 0.025 seconds and 40 filters to transform time-domain audio into rich spectrograms, effectively isolating the high-frequency crunching sounds of the weevil from background noise.

- Performance and Accuracy

The model achieves a strong overall validation accuracy of 88.8% and an F1 score of 0.88. It demonstrates perfect precision in identifying healthy coconut trees (100% accuracy) and successfully detects 69.3% of infected cases, minimizing false positives in the field.

- Edge Optimization

Designed strictly for embedded deployment, the inference engine is incredibly resource-efficient. It requires a peak RAM usage of only 19 KB and completes a full processing cycle in approximately 145 to 147 milliseconds, ensuring the system can run continuously on limited battery power without relying on cloud connectivity.

4 From Prototype to Final Product

4.1 Printed Circuit Board

- Breadboard Implementation - Initial proof-of-concept and component testing.

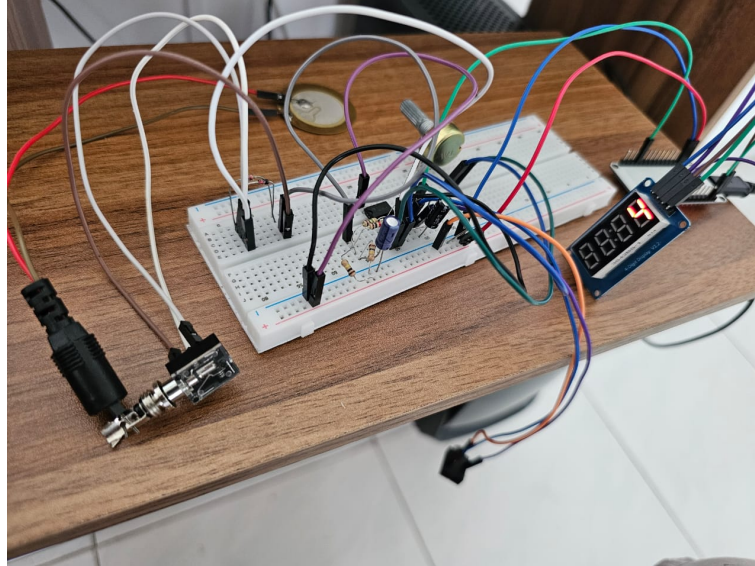


Figure 5: Solution Architecture

- Schematic Diagram - The logical mapping of the circuitry.

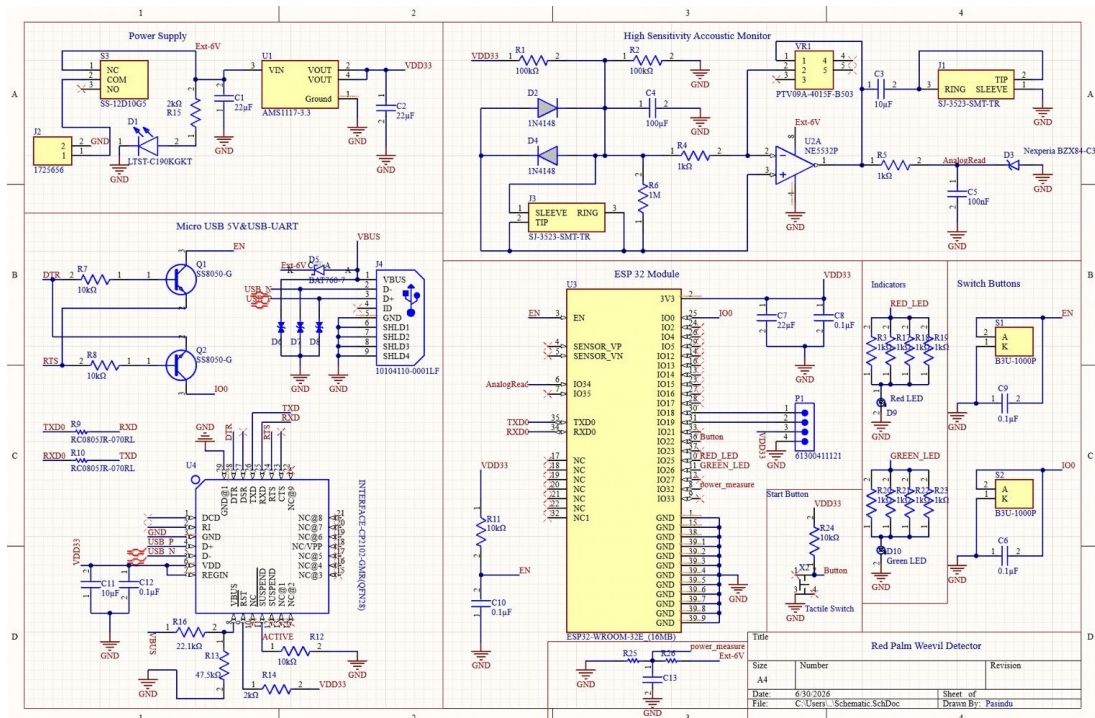
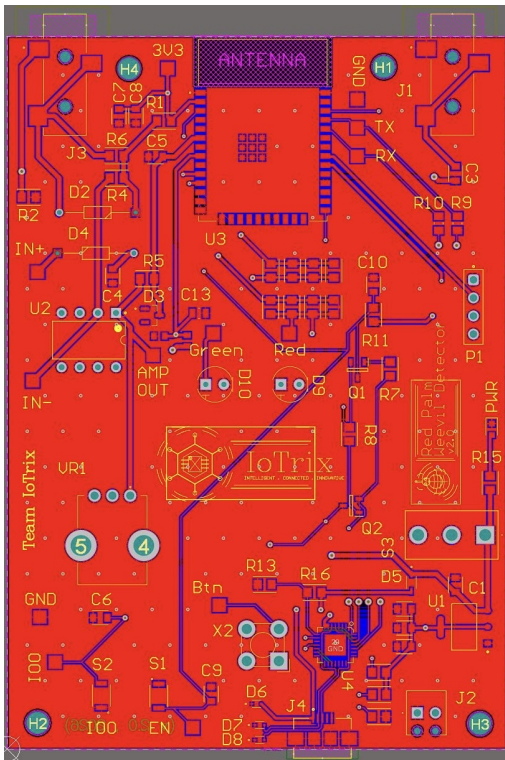
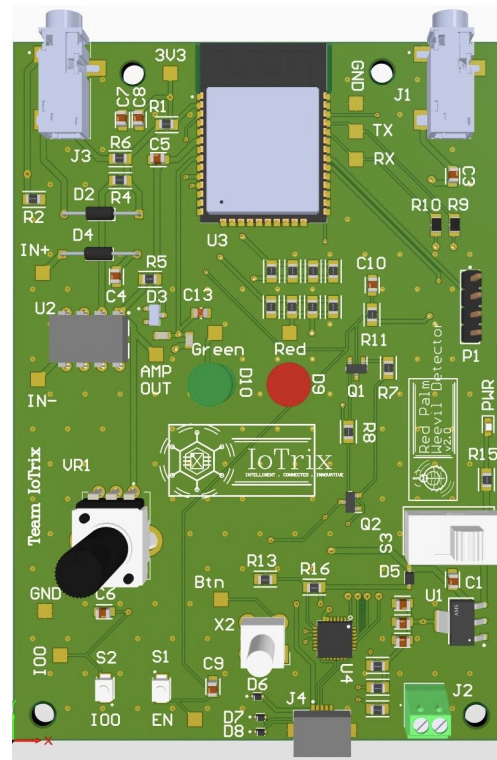


Figure 6: Schematic Diagram

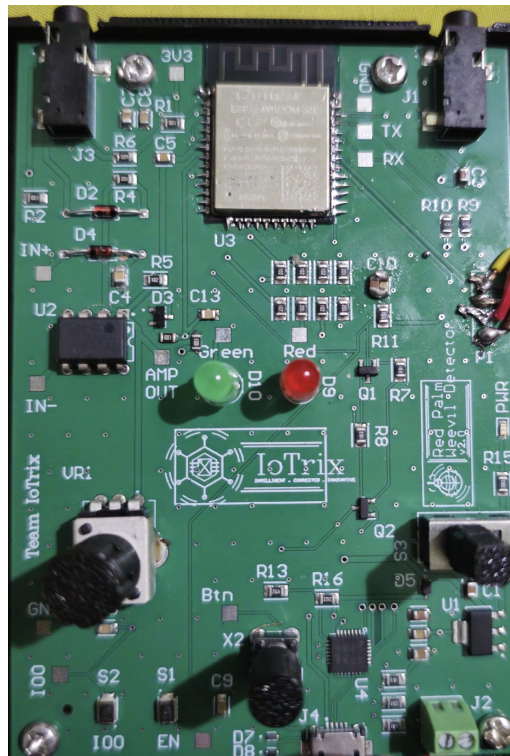
- PCB design - The layout and routing for the custom board.



(a) 2D Layout of the PCB



(b) 3D Layout of the PCB



(c) Physical PCB

4.2 Enclosure

- Ideation - Initial pencil sketches of the system's housing.

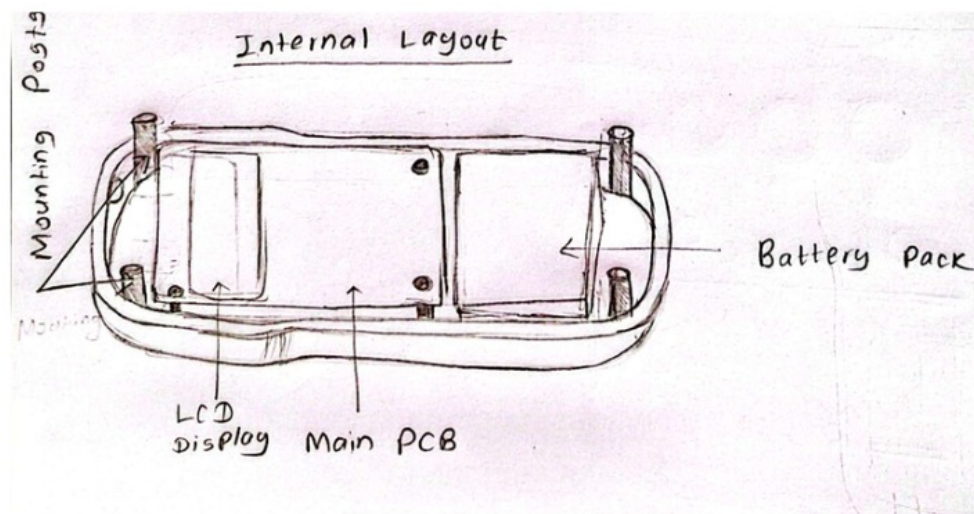
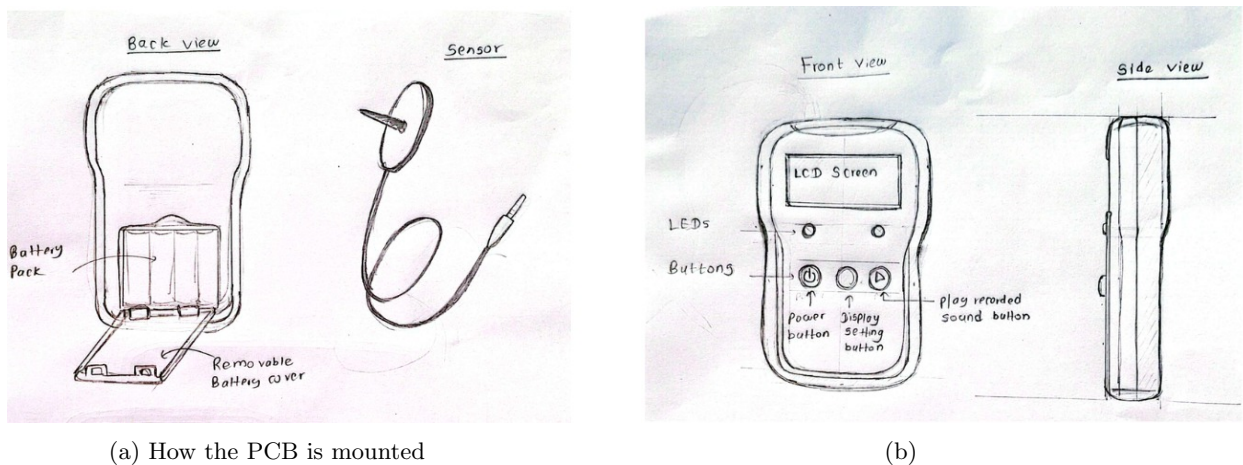


Figure 8



(a) How the PCB is mounted

(b)

Figure 9: Overall caption for both PCB sketches.

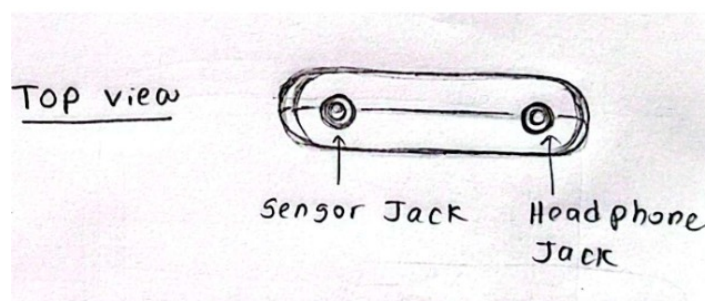
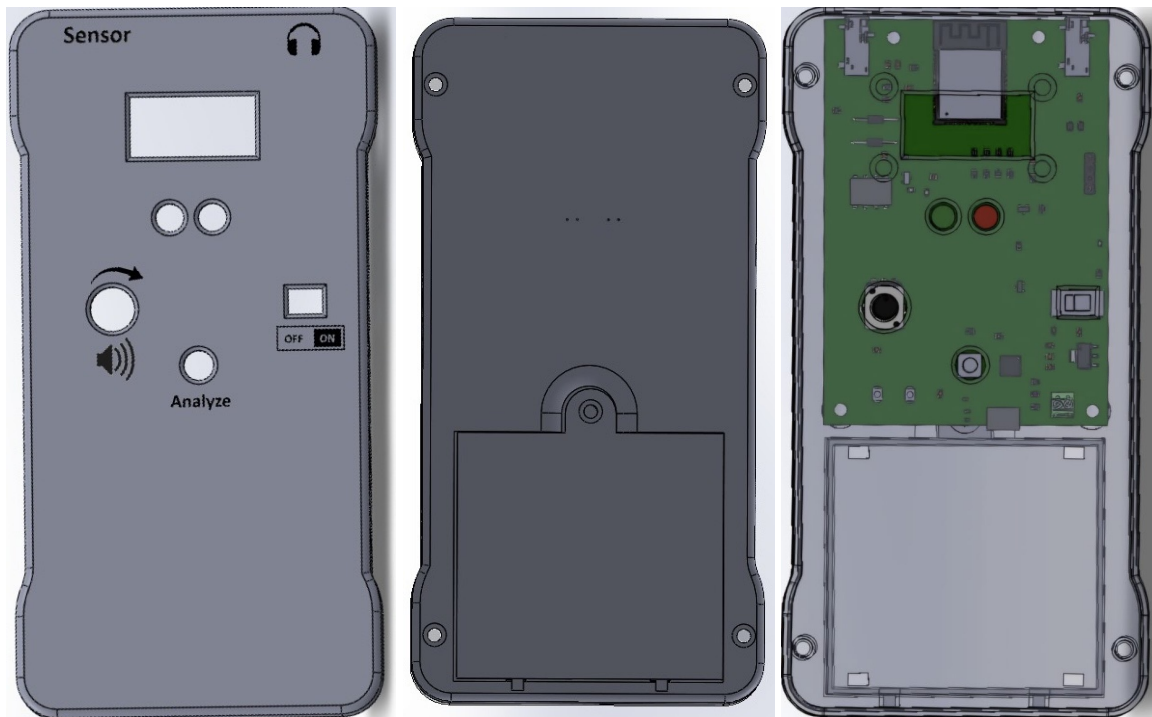


Figure 10

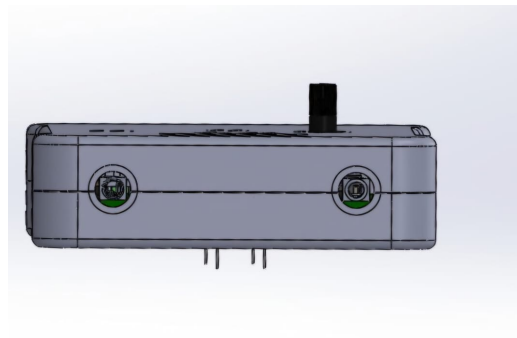
- CAD Modelling - SolidWorks designs, detailing how the acoustic sensors and PCB fit securely inside



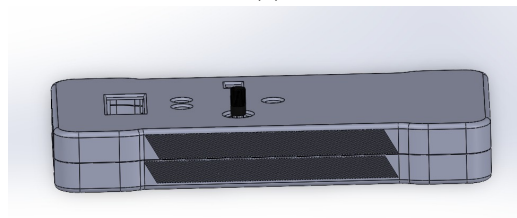
(a) Front-view

(b) Back-view

(c) How the PCB is mounted



(a)



(b)

- Final Fabrication - The printed enclosure ready for field deployment.



Figure 13: Final Enclosure after assembly

4.3 TinyML Model

- Data Collection
 - Collected 50 raw audio samples, split into a dataset of 500 segments for training.
 - Data was captured using the custom hardware sensor itself, rather than relying on open-source sound files, ensuring maximum accuracy for detecting the specific acoustic signatures of the weevil.



Figure 14: Recording Raw Audio Samples

- Training

Training was done using the commonly used platform for TinyML models, 'Edge Impulse'. An overall accuracy of 88.8% was achieved by the model when tested against the validation data set.

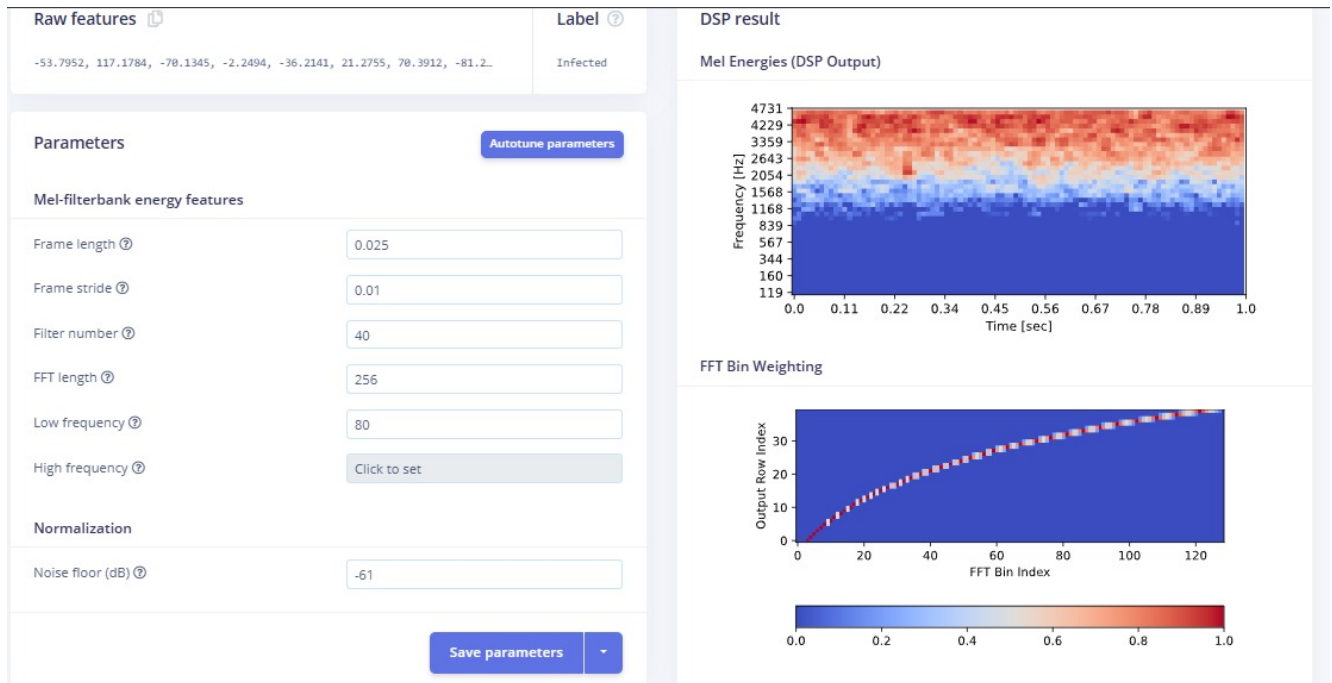


Figure 15

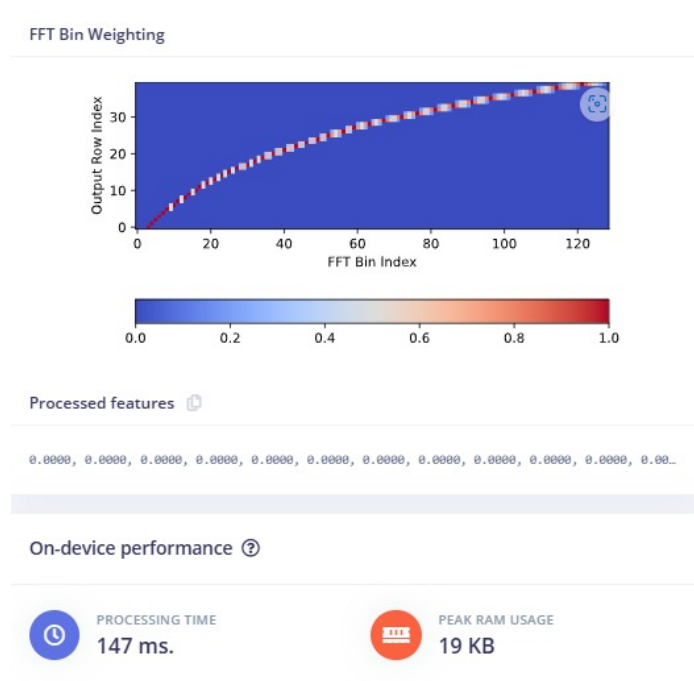


Figure 16

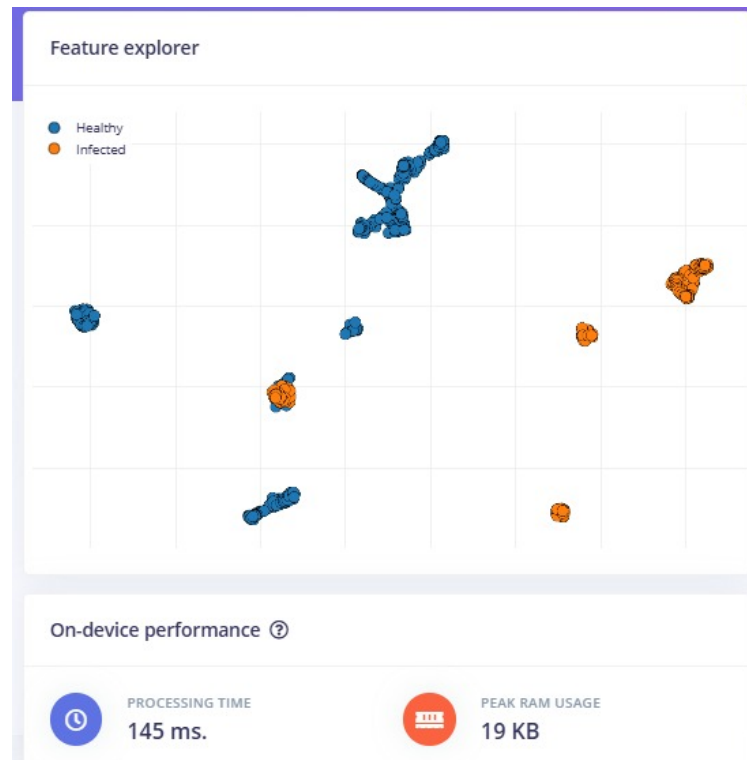


Figure 17

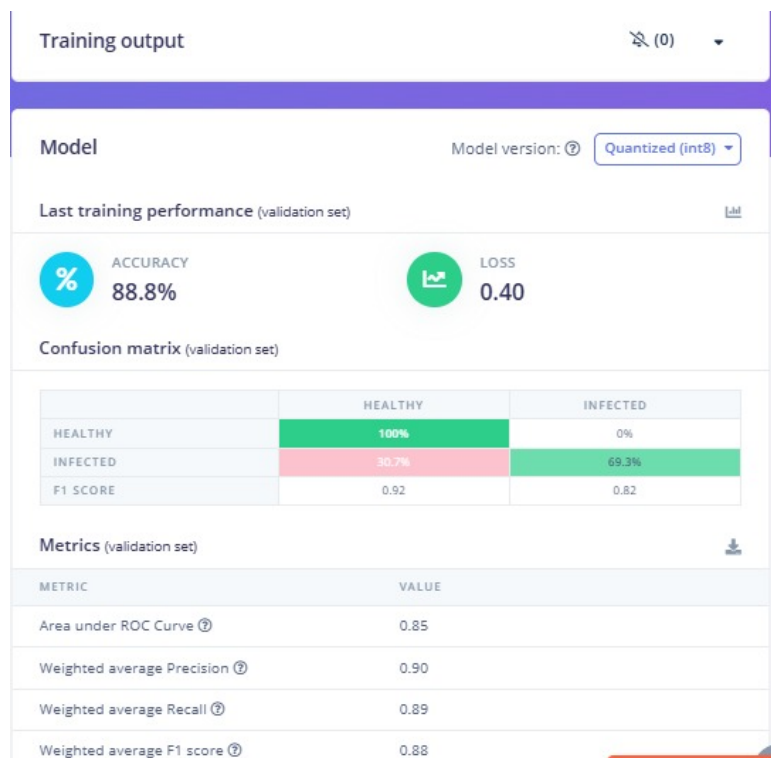


Figure 18

- Final Testing

Real-world validation conducted on live coconut palms at the Koshena Coconut Estate.



(a) Output reflects RPW internal activity



(b) Testing the device with infected coconut trees



Figure 20

5 Budget Analysis and Cost Estimation

Table 1: Estimated Project Cost Breakdown

Item	Unit cost(LKR)	Quantity	Total(LKR)
PCB Order(JLCPCB)	5836.8	5	29184.79
Components (LCSC)			12549.35
Tax (Components)			3095.72
TAX (PCB)			3948.75
7 - segment Display		1	220
Nuts and Inserts			390
AAA Batteries	120	4	480
Enclosure	2600	1	2600
Piezo Sensor Unit	1500	1	1500
Total Estimated Cost			53968.61

6 Team Contributions

- Sasmitha K. A. P
 - Designed the hardware setup of the circuit.
 - Contributed to the design and training of the ML model.
 - Designed the schematics and the PCB layout.
 - Contributed to circuit testing and implementation.
 - Participated in ideation, user research, and analysis.
- Kaluarachchi K. A. P. R.
 - Designed the PCB layout.
 - Carried out soldering and assembly.
 - Contributed to the design and training of the ML model.
 - Contributed to circuit testing.
 - Participated in ideation, user research, and analysis.
- Kothalawala D. H.
 - Designed the enclosure.
 - Contacted relevant resource persons to carry out testing.
 - Contributed to circuit testing and implementation.
 - Participated in ideation, user research, and analysis.
- Ranaweera R. K. M. D. T
 - Contributed to train the ML model.

- Designed the enclosure.
- Contacted relevant resource persons to carry out testing.
- Contributed to circuit testing and implementation.
- Participated in ideation, user research, and analysis.